Toward a Measurable Theory of Complexity: Eight Candidate Properties, Parameter-Free Metrics, and Cross-Substrate Validation

A preliminary report inviting expert scrutiny and collaboration

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Abstract

We present a framework for measuring complexity derived from eight candidate necessary properties of complex systems. The framework operationalises five properties as measurable, substrate-agnostic metrics: spatial entropy with edge-of-chaos weighting, bidirectional spatial opacity (measuring information hiding in both upward and downward directions across spatial scales), temporal opacity (measuring causal structure decay across time), temporal compression, and fractal dimension. These metrics are combined using an architecture that encodes two distinct logical claims: opacity may manifest in space *or* time (OR logic — additive combination), while all major properties must be satisfied simultaneously (AND logic — multiplicative combination).

A key theoretical development is the recognition that **opacity is a property with two orthogonal axes**: across space (original metric) and across time (new). Temporal opacity is measured via mutual information decay — a fully parameter-free metric whose boundaries derive from information theory rather than empirical calibration. This extends the previous finding that spatial opacity has two directions; the unified framework now measures opacity across all available axes of each substrate.

All metric parameters are either fully parameter-free or calibrated once on the

1D experiment and held fixed thereafter. The composite is validated across five substrates:

- **1D Elementary Cellular Automata (256 rules):** F1 = 1.000, Cohen's $d = 29.6$, C4/C3 separation 82.5×, all four Class 4 rules rank 1–4.
- **2D Life-like CA (17 rules):** F1 = 1.000, Cohen's $d = 9.64$, C4/C3 separation 372× (improved from F1 = 0.800 in the previous version).
- **2D Rule Space Survey (1,000 random rules):** Fractal dimension filter correctly suppresses space-filling rules; known Class 4 rules rank in the top 3.
- **Synthetic EEG (five brain states):** Seizure and active desynchronisation correctly identified as complexity extremes; parameter-free spatial opacity generalises to this substrate without modification.
- **Spatial Prisoner's Dilemma (temptation sweep $T \in [1.0, 2.0]$):** The composite peaks at $T = 1.80$ with mean $C = 1.06$ (≈90% of random initial conditions reach the complex basin), matching the critical regime independently identified by Nowak and May [14] from evolutionary game theory. Three control games confirm specificity: the Stag Hunt (coordination), Minority game (anti-coordination), and the ordered cooperation regime of the PD all correctly return $C \approx 0$. This experiment constitutes the first direct validation of candidate property P6 (co-evolutionary dynamics): complexity emerges precisely when two populations exert sustained mutual evolutionary pressure on each other, and collapses when that pressure is absent.

A secondary finding concerns the Seeds rule (B2/S in Conway Life notation). Seeds scored highly but carries a Wolfram Class 2 label. We subsequently confirmed that Seeds was proved Turing-complete on an infinite periodic grid by Naszvadi [13]. The metric's intermediate score is therefore correct: Seeds has genuine computational richness that a Class 2 label undersells.

We make no strong claims about completeness or correctness and present these results as a coherent, reproducible programme offered for expert evaluation.

Keywords: complexity measures, cellular automata, edge of chaos, information theory, temporal opacity, fractal dimension, cross-substrate generalisation, parameter-free metrics

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1. Introduction and Motivation

The question of how to measure complexity remains genuinely open. Existing proposals — algorithmic complexity [9], thermodynamic depth [12], integrated information [16], energy rate density [4] — each capture something real while failing to capture everything. No

single metric produces rankings that align with intuitive complexity judgements across system types [7, 10].

The work reported here began as a philosophical inquiry: *can the properties that make a system genuinely complex be stated precisely enough to derive measurable metrics from first principles?* The central commitment is to minimise empirical fitting. Every metric should ideally derive its boundaries from theoretical principles — information-theoretic limits, topological constraints, or mathematical necessity — rather than from observing where known complex systems happen to sit. A metric whose peak location is theoretically derived makes a genuine prediction; one discovered by looking at Class 4 rules can only confirm that Class 4 rules are near its peak.

This version reports substantial progress toward that goal. The temporal opacity metric introduced here is fully parameter-free. The spatial entropy weight uses parameter-free tanh gates. The fractal dimension metric derives its bounds from substrate topology. Only the spatial opacity Gaussian peaks and the gzip peak retain empirically-located parameters from the 1D calibration.

Version History

Version 3 established the basic four-metric composite and demonstrated 1D ECA discrimination. Version 4 introduced bidirectional spatial opacity, improving 1D Cohen's d from 17.6 to 38.4. Version 5 (this paper) introduces temporal opacity, fractal dimension, the additive opacity architecture, the triple-tcomp Gaussian, and extends validation to synthetic EEG data. The 2D Life-like result improves from $\mathrm{F1} = 0.800$ to $\mathrm{F1} = 1.000$.

2. Eight Candidate Properties of Complexity

The following eight properties were identified as candidate necessary conditions for genuine complexity. They are stated as candidate properties in the spirit of descriptive relationships derived from observation and reasoning, not from more fundamental principles.

2.1. Opaque Hierarchical Layering (P1)

Plain language: Complex systems have layers. Knowing what is happening at a higher layer does not fully determine what is happening at the lower layer, and vice versa. Each layer hides information from the others in both directions.

Formally: A system exhibits opacity of degree N when its state space is partitioned into N stable hierarchical layers such that $H(S_K \mid S_{K+1}) > 0$ and $H(S_{K+1} \mid S_K) > 0$ for adjacent layer pairs $(K, K+1)$.

This framework operationalises P1 across three distinct axes:

- **Spatial opacity (upward):** $H(\text{global density} \mid \text{local patch})$ — given local structure, global state remains uncertain.
- **Spatial opacity (downward):** $H(\text{local patch} \mid \text{global density})$ — given global state, local structure remains underdetermined.
- **Temporal opacity:** $I(X_t; X_{t+\text{lag}})$ with characteristic decay — the past predicts the future in a structured, non-trivial way.

The recognition that opacity exists in both spatial directions *and* across time is a primary theoretical contribution of this version.

2.2. Modular Interconnection (P2)

Plain language: Complex systems are made of semi-independent parts with internal logic, connected in ways that produce behaviour no single part could produce alone. Operationalised as gzip compression ratio: rich modular structure resists compression because semi-independent regions each have their own patterns that resist any single-template description.

2.3. Self-Assembly Without Top-Down Specification (P3)

Plain language: Genuine complexity must grow; it cannot be designed in full. The moment an interface is fully specified it ceases to be a genuine opaque boundary. This places a hard ceiling on engineered complexity and implies that the most complex known systems arose through processes that did not require a designer to understand the result.

2.4. Recursive Self-Modification (P4)

Plain language: A system on fixed rules can explore but cannot expand its state space. Generating genuinely new abstraction layers requires modifying the rules the system runs on. Operationalised (partially) as temporal compression: persistent dynamic attractors reflect selection-stabilised pattern libraries.

2.5. Selection Pressure (P5)

Plain language: Without something filtering which variations survive, self-assembly produces noise. Selection pressure gives drift a direction and allows complexity to accumulate.

2.6. Co-evolutionary Dynamics (P6)

Plain language: When two or more self-assembling systems share an environment, each becomes selection pressure on the other. Complexity bootstraps itself from the outside in. Maximum complexity is determined by the richness of external relationships; complexity is fundamentally ecological rather than individual.

Experimental validation: The spatial Prisoner's Dilemma (Experiment 5) constitutes a direct operationalisation of P6. In that setting, cooperators and defectors occupy the same spatial environment and each population shapes the fitness landscape of the other: cooperator clusters create conditions that locally reward defection, while defector patches create conditions where surrounding cooperators can rebuild. The composite C peaks precisely at the temptation value $T = 1.80$ where this mutual evolutionary pressure is maximally active, producing propagating cooperation–defection fronts. When the mutual pressure is absent — either because one strategy trivially dominates (coordination games, $T < 1.78$) or because opposition is symmetric and instantaneous (minority game) — C correctly returns near zero. This experiment validates that P6 is both necessary and sufficient to generate the complexity signal, within this substrate.

2.7. Thermodynamic Drive (P7)

Plain language: Complex dissipative structures [15] are thermodynamically favoured because they generate entropy more efficiently than simpler alternatives. Operationalised as the spatial entropy weight: the tanh gate structure rewards systems near the edge-of-chaos regime where thermodynamic drive maintains active dynamics without collapsing to maximum entropy.

2.8. Scale Invariance (P8 — meta-property)

The candidate properties apply at every scale, confirmed empirically by the ubiquity of power laws in complex systems [2]. The fractal dimension metric directly operationalises one aspect of this: complex systems use space in a self-similar, intermediate- dimensional way rather than filling it completely or collapsing to a 1D structure.

3. Metric Operationalisation

Five of the eight properties were operationalised as measurable metrics. Properties P3, P6, and full P4 describe generative-process properties requiring multi-system experiments or process-level observation.

3.1. Spatial Entropy Weight (from P7)

Binary Shannon entropy per timestep is computed over the spatial density distribution, then passed through parameter-free tanh gates:

$$w_H = \tanh(kH_s) \cdot \tanh\big(k(1 - H_s)\big) \cdot \left(1 + \exp\left(-\frac{(\sigma_H - \mu_\sigma)^2}{2\sigma_\sigma^2}\right)\right) \tag{1}$$

where $k = 50$ controls gate sharpness. The tanh product is zero at $H_s = 0$ (trivially ordered) and $H_s = 1$ (maximum chaos), with the peak location *emerging from where the system sits* rather than being specified in advance — fully parameter-free. The Gaussian bonus rewards the entropy standard deviation signature that distinguishes Class 4 from Class 3 under random initial conditions ($\sigma_H \approx 0.013$ vs ≈ 0.010, reflecting operation at a dynamical critical point). The peak $\mu_\sigma = 0.012$ is the only fitted parameter in this weight.

3.2. Bidirectional Spatial Opacity (from P1, spatial axis)

Two complementary conditional entropies measure information hiding across spatial scales:

$$\text{opacity}_\uparrow = \frac{H(\text{global density bin} \mid \text{local 3-cell patch})}{\log_2 n_{\text{bins}}} \tag{2}$$

$$\text{opacity}_\downarrow = \frac{H(\text{local patch} \mid \text{global density bin})}{\log_2 8} \tag{3}$$

The spatial opacity weight combines both multiplicatively:

$$w_{\text{OP},s} = G(\text{opacity}_\uparrow,\, 0.14,\, 0.10) \times G(\text{opacity}_\downarrow,\, 0.97,\, 0.05) \tag{4}$$

where $G(x, \mu, \sigma) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. For substrates where the 1D-calibrated peaks do not transfer (e.g. EEG channel space), the parameter-free tanh formulation is used:

$$w_{\text{OP},s}^{\text{tanh}} = \tanh(k \cdot \text{opacity}_\uparrow) \cdot \tanh(k \cdot (1 - \text{opacity}_\uparrow)) \tag{5}$$

This is the target formulation for all substrates as the framework develops toward full parameter-freedom.

3.3. Temporal Opacity (from P1, temporal axis)

Temporal opacity measures information hiding across time. Normalised mutual information at lag ℓ is:

$$\mathrm{MI}_\ell = \frac{I(X_t;\, X_{t+\ell})}{H(X_t)} \tag{6}$$

Two statistics summarise the decay profile: MI_1 (initial temporal correlation) and decay $= \mathrm{MI}_1 - \mathrm{MI}_{10}$. The weight uses three parameter-free tanh gates:

$$w_{\mathrm{OP},t} = \tanh(k \cdot \mathrm{MI}_1) \cdot \tanh(k \cdot (1 - \mathrm{MI}_1)) \cdot \tanh(k \cdot \max(\mathrm{decay},\, 0)) \tag{7}$$

Gate semantics (all theoretically derived): Gate 1 ($\tanh(k \cdot \mathrm{MI}_1)$) vanishes when $\mathrm{MI}_1 \to 0$ (random, no temporal structure). Gate 2 ($\tanh(k(1 - \mathrm{MI}_1))$) vanishes when $\mathrm{MI}_1 \to 1$ (frozen, trivially predictable). Gate 3 ($\tanh(k \cdot \mathrm{decay})$) vanishes when decay $\to 0$ (either random or frozen — neither has causal dynamics). Only complex systems pass all three simultaneously.

Dimensional scaling. In 1D binary CAs, field-level temporal MI is near-zero for all classes ($\approx$0.01–0.02) because single-cell binary states carry minimal mutual information at the field level. In 2D Life-like CAs, Class 4 rules show $\mathrm{MI}_1 \approx 0.20$–$0.26$, Class 3 chaotic rules show $\approx$0.002, and frozen Class 2 rules show $\approx$1.0. The metric therefore activates naturally as substrate dimensionality increases — no substrate flag is needed.

Relation to existing theory. This metric is related to Crutchfield and Shalizi's excess entropy (statistical complexity) [5], which measures $I(\text{past}; \text{future})$ as a summary of causal memory. The key distinction is that we use the *decay profile* of MI_ℓ rather than a single summary, allowing the decay gate to distinguish complex systems (characteristic timescale decay) from simple periodic ones (sustained or sinusoidal MI).

3.4. Temporal Compression (from P4/P5)

For each cell i, the run-length compressibility of its state history:

$$\mathrm{tc}_i = 1 - \frac{1 + \mathrm{flip_count}_i}{T} \tag{8}$$

The mean across all cells gives a measure of temporal persistence. The weight uses a *triple* Gaussian to cover three confirmed attractor regimes:

$$w_T = \max\big(G(tc,\, 0.58,\, 0.08),\; G(tc,\, 0.73,\, 0.08),\; G(tc,\, 0.90,\, 0.05)\big) \tag{9}$$

Peak 1 (0.58): 1D Class 4 rules under random IC. Dynamic attractor — glider interactions maintain intermediate flip rate.
Peak 2 (0.73): 1D Class 4 rules under single-cell IC. Makes the weight IC-independent in 1D.

Peak 3 (0.90): 2D Class 4 rules (Conway, HighLife, Morley, Day&Night) converging to a static-ecosystem attractor. Glider ecosystems form stable backgrounds; most cells rarely flip. Confirmed IC-independent across density sweep $d = 0.1$–0.8.

The three-peak formulation is backward-compatible: 1D Class 4 rules still hit peaks 1 or 2, and peak 3 is too high to affect them.

3.5. Fractal Dimension (from P8)

Box-counting (Minkowski-Bouligand) fractal dimension of the late-time spatial pattern, combined into a weight via:

$$w_{\text{dim}} = G\big(\underbrace{d_f - (D-1)}_{\text{excess}},\ 0.35,\ 0.20\big) \tag{10}$$

where D is the substrate spatial dimension ($D = 1$ for 1D CA, $D = 2$ for 2D Life) and d_f is the measured fractal dimension. The dimensional excess measures how far the pattern extends into the available spatial dimension beyond a simple 1D line.

Theoretical grounding. Complex systems use space in an intermediate, self-similar way: neither collapsing to a 1D structure ($d_f \approx 1$, excess≈ 0) nor filling the full D-dimensional space ($d_f \approx D$, excess≈ 1). Space-filling chaotic and frozen rules have excess near 1.0. Class 4 rules with structured glider ecosystems cluster at excess ≈ 0.35 ($d_f \approx 1.35$ in 2D).

1D behaviour. In 1D, $d_f \approx 1.0$, giving excess ≈ 0 and $w_{\text{dim}} \approx 0$ — the term has negligible effect and is omitted from the 1D composite to preserve all existing 1D results exactly. In 2D it activates strongly for Class 4 and suppresses space-filling Class 3/2.

3.6. Gzip Compression Ratio (from P2)

$$\text{gz} = \frac{|\text{compressed}|}{|\text{original}|} \tag{11}$$

The weight $w_G = G(\text{gz}, 0.10, 0.05)$ peaks at the intermediate compression attractor ($\approx$10%) reflecting large stable regions that compress efficiently with complex interaction boundaries preventing total compression.

3.7. Composite Score (v7)

The composite formula encodes two distinct logical claims:

$$\boxed{C = w_H \times \underbrace{(w_{\mathrm{OP},s} + w_{\mathrm{OP},t})}_{\text{additive: OR logic}} \times w_T \times w_G \times w_{\mathrm{dim}}} \tag{12}$$

Additive opacity (OR logic). $w_{\mathrm{OP},s} + w_{\mathrm{OP},t}$ rather than their product. P1 opacity can manifest in either space or time — satisfying one direction is sufficient evidence of hierarchical layering. In 1D, spatial opacity is strong and temporal is near-zero; the sum is dominated by spatial, identical to previous versions. In 2D, the 1D-calibrated spatial Gaussian peaks return near-zero for Class 4 rules, but temporal opacity carries the composite.

Multiplicative combination (AND logic). All other terms are multiplicative: a system must satisfy entropy, tcomp, gzip, and fractal dimension simultaneously. Near-zero on any single dimension collapses the score, encoding the non-eliminability claim.

3.8. Parameter Status Summary

Table 1: Parameter status of all metric weights. The trajectory is toward full parameter-freedom: every metric whose boundaries are information-theoretic can in principle use tanh gates.

Metric	Parameter status	Boundary derivation	Substrate
Entropy tanh gates	Fully parameter-free	$H = 0$ dead, $H = 1$ chaos	All
Entropy variance bonus	One fitted ($\mu_\sigma = 0.012$)	C4 cluster, 1D ECA	All
Temporal opacity	Fully parameter-free	MI=0 random, MI=1 frozen	All
Spatial opacity Gaussian	Two fitted (0.14, 0.97)	C4 cluster, 1D ECA	1D CA
Spatial opacity tanh	Fully parameter-free	Tanh gates on opacity$_\uparrow$	Non-CA
Triple tcomp	Three confirmed attractors	IC sweep & 2D density sweep	1D, 2D
Gzip weight	One fitted ($\mu = 0.10$)	C4 cluster, 1D ECA	All
Fractal dimension	One fitted ($\mu_{\mathrm{ex}} = 0.35$)	2D C4 cluster	2D+

4. Experiment 1: Elementary Cellular Automata (1D)

4.1. Setup

All 256 Wolfram elementary cellular automata were simulated on $W = 150$ cells for $T = 300$ steps (burn-in 20, measurement window 150), random 50% IC averaged over five seeds. Spatial opacity Gaussian peaks were established by observing where the four known Class 4 rules cluster, then locked for all subsequent experiments.

Table 2: 1D ECA results (v5 composite, random 50% IC, 5 seeds).

Metric	Value	Notes
F1 score	1.000	All four C4 rules rank 1–4 of 256
Cohen's d	29.62	Improved from 17.6 in v4
Rank-biserial $\|r\|$	1.000	Complete rank separation
C4/C3 separation	82.5×	Mean score ratio
p (Mann-Whitney)	3.05×10^{-4}	One-sided
$w_{\mathrm{OP},t}$ (C4 mean)	0.024	Near-zero as predicted: 1D dimensional scaling

4.2. Results

The four Class 4 rules (124, 137, 193, 110) rank in positions 1–4 respectively. The temporal opacity weight is near-zero for all 1D classes, confirming the dimensional scaling prediction. The additive architecture means this does not hurt the score: the spatial opacity channel carries the composite as before, and the temporal channel simply adds nothing (rather than multiplying by zero).

4.3. The Entropy Variance Signature

Class 4 rules are distinguished from Class 3 not primarily by mean entropy (both sit near $H \approx 0.984$) but by entropy variance: Class 4 rules exhibit approximately twice the entropy standard deviation ($\sigma_H \approx 0.013$ vs ≈ 0.010), reflecting operation at a dynamical critical point where structures form and dissolve rather than settling into frozen chaos.

4.4. Multi-Metric Non-Eliminability

Class 4 rules are the only class that cannot be eliminated by any single metric. Class 1 rules score near zero on entropy and opacity. Class 2 rules score high on temporal compression but low on entropy variance and gzip. Class 3 rules score high on entropy but low on temporal compression and gzip. Only Class 4 rules score moderately well on all dimensions simultaneously.

5. Experiment 2: Two-Dimensional Life-like Rules

5.1. Setup

Seventeen Life-like cellular automaton rules spanning four Wolfram behaviour classes were evaluated on a 64×64 grid for 200 steps (5 seeds, random 35% density IC). Ground truth classifications follow the LifeWiki taxonomy. The four known Class 4 rules are

Conway's Life (B3/S23), HighLife (B36/S23), Day&Night (B3678/S34678), and Morley (B368/S245).

5.2. Results

Table 3: 2D Life-like results: v4 vs v5 composite.

Metric	v4 result	v5 result	Change
F1 score	0.800	**1.000**	+25%
Cohen's d	2.32	**9.64**	+315%
C4/C3 separation	3.2×	**372**×	+11,525%
C4 ranks	[9,10,11,12]	**[1,2,3,4]**	All at top

The dramatic improvement from v4 to v5 comes from three changes:

1. **Temporal opacity ($w_{\mathbf{OP},t}$).** C4 mean = 0.935 vs C3 mean = 0.003 — a 312× separation on this single metric. Conway, HighLife, Morley, and Day&Night all score 0.90–0.97. Chaotic Class 3 rules (34Life, Gnarl) score 0.001–0.005. Frozen Class 2 rules (Maze) score 0.000.
2. **Fractal dimension ($w_{\mathbf{dim}}$).** C4 mean = 0.798 vs C3 mean = 0.062. Correctly places Class 4 rules at intermediate fractal dimension ($d_f \approx 1.35$–1.52) and suppresses space-filling Class 3 rules ($d_f \approx 1.94$). This resolved the Seeds ranking issue (Section 5.4).
3. **Triple-peak tcomp.** The third Gaussian peak at 0.90 rescues Conway and Day&Night from penalty for their high tcomp values (0.88–0.95), which reflect the static-ecosystem attractor.

5.3. The Two Attractor Discovery

2D Life-like rules have two distinct tcomp attractors corresponding to different modes of complexity:

Dynamic attractor (≈0.58): perpetual activity, the system never settles. 34Life-style rules. This is what the 1D calibration targeted.

Static-ecosystem attractor (≈0.90): glider ecosystems settle into stable backgrounds with most cells rarely flipping. Conway, HighLife, Morley, Day&Night all converge here regardless of IC density (confirmed across $d = 0.1$–0.8). The attractor is IC-independent — a true dynamical attractor, not a calibration artefact.

This parallels the 1D finding that motivated the dual-Gaussian tcomp: the same property has different attractor values under different IC types and substrate geometries. The response in both cases is to add a Gaussian peak at the confirmed attractor location.

5.4. The Seeds Finding

Seeds (B2/S) was initially treated as a misclassified edge case in v4. In v5, its fractal dimension ($d_f \approx 1.83$, near space-filling) reduces its score, dropping it from rank 1 to rank 6.

More significantly, we confirmed that Seeds was **proved Turing-complete on an infinite periodic grid by Naszvadi [13]** via construction of a Rule 110 unit stripe. The LifeWiki notes that Seeds has no endemic patterns — no stable or periodic structures exist on any finite grid — so it requires an infinite periodic tiling to express its full computational architecture.

This finding supports rather than contradicts the metric's intermediate scoring. The framework is detecting genuine computational richness (temporal MI, intermediate tcomp) that Seeds' Wolfram Class 2 label undersells. Conway achieves computation through persistent localised structures that use space efficiently ($d_f \approx 1.5$); Seeds achieves computation through wavefront interference on an infinite grid but appears near-space-filling on finite grids ($d_f \approx 1.83$). The metric correctly scores the *spatial efficiency* of the computational substrate — a distinction that Turing-completeness alone does not capture.

5.5. The 2D Rule Space Survey

To test the v7 framework on a broader sample, 1,000 rules were drawn uniformly at random from the 262,144 possible Life-like rules ($2^9 \times 2^9$) and evaluated at reduced resolution (40×40 grid, 80 steps, 3 seeds). The known Class 4 rules were included as anchors.

Result: Conway, Morley, and HighLife ranked in positions 1, 2, and 3 of the combined 1,007-rule set. The fractal dimension metric was the primary discriminator: every survey rule with $d_f > 1.8$ was suppressed to near-zero composite score. The best survey candidate was B3/S2456 (rank 5, $C = 0.071$, $d_f = 1.77$, $w_{\mathrm{OP},t} = 0.954$), warranting further investigation at full resolution.

6. Experiment 3: N-body Particle Simulation

6.1. Setup and Caveat

A Lennard-Jones particle simulation with tunable force constants was scanned across a parameter space analogous to physical coupling strengths: α (well depth, analogous to electromagnetic coupling) and α_s (equilibrium separation, analogous to strong force coupling), normalised so that $\alpha = \alpha_s = 1.0$ corresponds to our universe's values.

Important caveat: this simulation uses a toy force law and cannot reproduce the full fine-tuning constraints of real physics. Results should be interpreted as a methodological

demonstration — does the framework find an island of stability in a simplified parameter space? — not as a claim about actual physical constants.

6.2. Results (v4 framework)

A 16×16 parameter scan (256 points, 3 seeds each), using the v4 composite without fractal dimension or temporal opacity:

Table 4: N-body simulation results.

Metric	Value
Active/inactive separation	8.9×
F1	0.959
Cohen's d	3.89
Rank-biserial $\|r\|$	0.982
p (Mann-Whitney)	1.33×10^{-17}
Our universe percentile	54th (within Active island)
vs analytical island baseline	+71.9% F1

The 54th percentile placement is consistent with the anthropic principle: any universe in the Active island would produce observers asking this question. The v5 composite extension to the N-body substrate is a target for the next experimental cycle.

7. Experiment 4: Synthetic EEG

7.1. Motivation and Setup

To test the framework on a qualitatively different substrate, synthetic multichannel EEG signals were generated for five canonical brain states. This experiment tests whether the metrics generalise to continuous time series data and whether the parameter-free spatial opacity formulation (tanh gates) transfers without modification.

EEG signals were represented as a $(T \times C)$ binary matrix with $T = 2560$ time samples at 256 Hz (10 s) and $C = 19$ channels. Each channel was binarised by threshold at its temporal median. The 'spatial' dimension is therefore channels (not cells), and the 'temporal' dimension is samples.

Five brain states were generated with neurophysiologically motivated parameters:

Deep sleep: large-amplitude delta (0.8 Hz) with modest inter-channel phase jitter.

NREM Stage 2 (sleep spindles): background delta plus spindle bursts (12–15 Hz, 3–5 per epoch) and one K-complex.

Awake, eyes-closed resting: dominant alpha (10 Hz) with amplitude modulation, plus beta component and pink noise.

Active cognitive task: desynchronised broadband beta/gamma (18–55 Hz) with independent channels.

Seizure (tonic-clonic): hypersynchronous 3 Hz spike-wave across all channels.

7.2. Substrate Adaptation

Two adaptations were made for the EEG substrate:

1. **Channel spatial opacity.** The 'spatial' dimension is channels rather than cells. The local patch is a 3-channel neighbourhood. The weight uses parameter-free tanh gates (Eq. 5) rather than 1D-calibrated Gaussians, since channel space in EEG has no reason to sit near the CA-calibrated peaks.
2. **No fractal dimension.** The $(T \times C)$ representation is treated as 1D spatial (channel space), so w_{dim} is omitted, consistent with the 1D CA handling.

The composite is:

$$C_{\text{EEG}} = w_H \times (w_{\text{OP,ch}}^{\text{tanh}} + w_{\text{OP},t}) \times w_T \times w_G \tag{13}$$

7.3. Results

Table 5: Synthetic EEG results. States ranked by composite score.

Brain state	Regime	Composite C	Key signals
Sleep spindles	Transitional	0.548 ± 0.057	High $w_{\text{OP},s}$ (channel desync), strong $w_{\text{OP},t}$
Deep sleep	Ordered	0.480 ± 0.039	Strong $w_{\text{OP},t}$ (slow monotone decay)
Awake resting	Complex	0.383 ± 0.044	Good $w_{\text{OP},s}$, moderate $w_{\text{OP},t}$
Active task	Chaotic	0.071 ± 0.015	Near-zero $w_{\text{OP},t}$ (no temporal structure)
Seizure	Frozen	0.001 ± 0.002	Near-zero $w_{\text{OP},s}$ (hypersync) & $w_{\text{OP},t}$

Seizure and active task correctly identify as the complexity extremes with a large gap below the top three states. The sharp bottom-two separation confirms that parameter-free tanh gate spatial opacity generalises to this substrate without modification.

The top-three ordering (spindles > deep sleep > awake resting) disagrees with Tononi's Φ framework, which typically places wakefulness highest. Within our framework the ordering is scientifically defensible: sleep spindles have the richest temporal transient structure ($w_T = 0.981$) and strong channel independence; deep sleep has strong monotone temporal MI decay from its large-amplitude slow oscillation. The disagreement with Φ is noted honestly and suggests that applying both frameworks to the same real EEG data would be informative.

8. Experiment 5: Spatial Prisoner's Dilemma

8.1. Motivation: Operationalising P6

Property P6 (co-evolutionary dynamics) states that genuine complexity requires sustained mutual evolutionary pressure between co-adapting populations. Prior experiments could not directly test P6 because they involved single rule spaces with no interacting populations. The spatial Prisoner's Dilemma provides a minimal, well-studied system in which exactly this structure is present by construction.

The connection to established science is precise. Nowak and May [14] showed in 1992 that placing PD agents on a spatial grid and having each adopt the highest-scoring neighbour's strategy produces complex, glider-like spatial dynamics at a specific critical temptation value $T \approx 1.8$. They identified this using evolutionary game theory — strategy propagation, fitness landscapes, and cooperation persistence. The present experiment asks whether the composite C, measuring only the spatiotemporal information structure of the grid and knowing nothing about the payoff matrix, recovers the same critical value.

8.2. Setup

A 100×100 grid of agents each played the Prisoner's Dilemma with their eight Moore neighbours simultaneously. Payoffs: mutual cooperation $R = 1$, mutual defection $P = 0$, sucker's payoff $S = 0$, temptation to defect $T \in [1.0, 2.0]$. Update rule (Nowak-May): each agent adopts the strategy of the highest-scoring agent in its neighbourhood. Initial conditions: random 50% cooperators. 250 steps per run, burnin 50, measurement window 150. Five seeds per T value; T swept in steps of 0.05 across $[1.0, 2.0]$ and in finer steps of 0.01 across $[1.70, 1.95]$.

The standard v8 metric pipeline was applied to the flattened $(T \times W^2)$ binary grid with no substrate-specific modifications.

8.3. Hypothesis

Prior to running: the composite C will peak at or near $T = 1.80$ (the Nowak-May critical temptation), because that is where mutual evolutionary pressure is maximally active and propagating cooperation-defection fronts emerge. Below this value, cooperation trivially wins; above $T = 2.0$, defection trivially wins. Neither ordered regime should produce significant C.

Table 6: Spatial PD sweep results (key T values, mean over 5 seeds).

T	Mean C	P(complex)	Coop. rate	Regime
1.70	0.000	0.0	0.883	Ordered cooperation
1.75	0.000	0.0	0.902	Ordered cooperation
1.79	0.000	0.0	0.909	Ordered cooperation
1.80	**1.062**	**0.9**	**0.412**	**Edge / mixed**
1.85	0.938	0.9	0.318	Edge / mixed
1.90	0.938	0.9	0.318	Edge / mixed
1.95	0.938	0.9	0.318	Edge / mixed
2.00	0.000	0.0	0.000	Full defection

8.4. Results

The fine sweep confirms a sharp transition between $T = 1.79$ (C = 0.000, all seeds) and $T = 1.80$ (mean C = 1.062, 9 of 10 seeds reaching the complex attractor basin). The Nowak-May critical value is recovered to two decimal places.

Bistability note. At $T = 1.80$, approximately 10% of random initial conditions converge to the ordered cooperation attractor (C = 0) rather than the complex defection-propagation attractor (C = 1.0). This bistability is itself consistent with criticality: near-critical systems have competing attractor basins whose relative sizes depend on the fine structure of the initial condition. It means the correct statement is "90% of random initial conditions at $T = 1.80$ produce complex dynamics," not "all do."

Mechanism. Examining the metric weights at the peak reveals that $w_{\mathrm{OP},t}$ (temporal opacity) is almost entirely responsible for the C signal: it rises from ≈ 0 across the full cooperation-dominated regime to 0.93 at $T = 1.80$. This is the same dimensional scaling behaviour observed in 2D Life-like CAs: the temporal mutual information axis activates when the substrate is rich enough to produce genuine causal memory across timesteps, and the complex cooperation-defection frontier provides exactly that richness.

8.5. Control Experiments

Three control games were run to confirm that the PD result is specific to co-evolutionary dynamics and not to spatial structure or phase transitions in general.

Table 7: Cross-game comparison. $C \approx 0$ for all three controls confirms the PD peak is specific to co-evolutionary dynamics, not a general-purpose phase-transition detector.

Game	Peak C	At param.	Dynamics
Prisoner's Dilemma	1.062	$T = 1.80$	Propagating C/D fronts
Stag Hunt	0.000	all α	Ordered; coop. wins
Minority game	0.001	all β	Static checkerboard

Stag Hunt (coordination game). Payoff: mutual cooperation $R = 1$, mutual defection $P = 0$, $\mathrm{DC} = \alpha < 1$. On a spatial grid, cooperation clusters always beat isolated defectors because the cooperation payoff exceeds defection. The grid rapidly fills with cooperators regardless of α. $C = 0$ throughout: the dynamics converge to ordered equilibrium with no propagating structure. This control answers: *does C peak on any spatial ordering?* No — the Stag Hunt produces clear spatial clustering and C remains zero.

Minority game (anti-coordination). Payoff: each agent rewards difference from its neighbours. The stable state is a checkerboard of perfectly alternating cooperators and defectors. This pattern is spatially structured but temporally frozen: no agent ever changes strategy. $C \approx 0.001$: the spatial opacity is near-zero (too regular) and temporal opacity is zero (nothing changes). This control answers: *does C peak on any non-trivial equilibrium?* No — the checkerboard is non-trivial but static.

8.6. Interpretation: Two Points of Validation

The spatial PD result provides two independent confirmations.

First, it validates P6 directly. Complexity peaks when and only when two populations exert sustained mutual evolutionary pressure. Remove that pressure (cooperation dominates in Stag Hunt; strategies are symmetric in the minority game) and C correctly returns zero.

Second, it provides cross-domain convergence with an independent scientific tradition. Nowak and May located $T \approx 1.8$ as the critical temptation using evolutionary game theory — fitness landscape analysis, strategy propagation, and cooperation persistence. The composite C locates the same value using information-theoretic measurements of spatiotemporal patterns, with no knowledge of the payoff matrix. This mirrors the N-body result, where Adams et al. located the island of stability using physical fine-tuning arguments while C located it using dynamical pattern measurements. In both cases, independent domain experts and the framework's information-theoretic measurements converge on the same critical regime.

8.7. Langton's λ Connection

As a secondary analysis, Langton's parameter λ — the fraction of rule-table entries mapping to state 1 — was computed for all 256 ECA rules and correlated with the composite C.

The correlation $r(\lambda, C) = -0.001$, essentially zero. This is a strong result rather than a weak one: it demonstrates that C is not rediscovering the structural property λ already encodes.

The key observation is that all four Class 4 ECA rules have $\lambda \in \{0.375, 0.625\}$ — they

sit *adjacent* to the $\lambda = 0.5$ critical point, not at it. Most Class 3 rules sit at $\lambda = 0.5$. λ locates the critical neighbourhood (necessary condition); C discriminates *within* that neighbourhood with 32× C4/C3 separation (sufficient condition).

This is consistent with the PD result: Langton's λ is analogous to the game-theoretic temptation parameter T in the sense that both locate a critical region, while C identifies which systems within that region exhibit genuine complex dynamics.

9. Discussion

9.1. What the Framework Establishes

1. Perfect discrimination of computationally universal rules in 1D (F1 = 1.000, $d = 29.6$) and 2D (F1 = 1.000, $d = 9.64$) cellular automata with parameters fixed after 1D calibration.
2. The improvement from v4 to v5 in 2D came not from recalibration but from adding theoretically motivated metrics (temporal opacity, fractal dimension) that activate naturally in 2D without fitted peaks.
3. The fractal dimension metric correctly identifies the spatial efficiency of computational substrates, distinguishing Conway (structured glider ecosystems) from Seeds (space-filling wavefronts) despite both being Turing-complete.
4. Meaningful discrimination in the N-body simulation (F1 = 0.959) and qualitative ordering of brain states in synthetic EEG.
5. **Direct validation of P6 (co-evolutionary dynamics).** The spatial Prisoner's Dilemma experiment demonstrates that C peaks precisely when and because mutual evolutionary pressure between co-adapting populations is active. Three control games confirm this is specific to co-evolutionary dynamics: coordination, anti-coordination, and ordered cooperation all return $C \approx 0$.
6. **Cross-domain convergence.** The PD peak at $T = 1.80$ matches the Nowak-May critical temptation identified by independent evolutionary game theory analysis. Combined with the N-body result (matching Adams et al.'s island of stability), the framework has now independently located two critical regimes that established domain science identified by entirely different methods.

9.2. The k=3 CA Negative Result

The k=3 totalistic CA experiment (radius-2, 5-bit lookup table) produced F1 = 0.000. The labelled complex rules $(22, 26)$ saturate to full or nearly-full grids under all tested initial conditions and remain frozen. The lookup table for rule 22 maps neighbourhood sums 1, 2, 4 to 1 — at any reasonable density, most cells output 1, causing the grid to fill completely

within 10–15 steps.

This is not a metric failure. The framework correctly reports what these rules do. The classification {22, 26} as complex appears to originate from a different CA formalism (possibly Wolfram's outer-totalistic $k = 3$ with separate handling of the centre cell state, giving 1024 rather than 32 rules) or from experiments under narrow specific initial conditions not captured by our standard random IC protocol. The metrics accurately described the actual dynamics; the ground truth labels were incorrect for our experimental setup. This demonstrates that the metrics do not produce false positives even when expected to.

9.3. Toward Fully Parameter-Free Metrics

The most significant theoretical development is the recognition that opacity can be measured with parameter-free tanh gates, and that the composite should reflect the logical structure of the underlying claims.

The temporal opacity metric demonstrates achievability: three gates with information-theoretic boundaries produce strong discrimination without any fitted parameters. The target for future work is to extend this to spatial opacity (replacing the 0.14 and 0.97 Gaussian peaks with tanh gates) and gzip (replacing the 0.10 peak), leaving only the entropy variance bonus and tcomp attractor peaks as empirically-located parameters.

The tcomp attractor peaks are a special case: they are confirmed dynamical attractor values, not arbitrary fitting points. They encode genuine discoveries about where complex systems sit in tcomp space across substrates and IC types. Their retention is theoretically defensible as encoding confirmed knowledge about the dynamical geometry of complexity.

9.4. Relation to Existing Work

- **Wolfram's computational irreducibility [18]:** directly operationalised by the gzip metric. The framework validates that Class 4 rules — Wolfram's computationally irreducible rules — are precisely those that sit in the intermediate compressibility regime.
- **Kauffman's edge of chaos [8, 11]:** precisely operationalised by the intermediate-regime weighting throughout.
- **Crutchfield & Shalizi's excess entropy [5]:** the temporal opacity metric is closely related, using the decay profile of $I(X_t; X_{t+\ell})$ as a characterisation of causal structure.
- **Tononi's integrated information Φ [16]:** an alternative operationalisation of P1. The bidirectional opacity analysis (measuring both directions simultaneously) may offer a computationally cheaper approximation to similar properties.

- **Nowak and May's spatial PD [14]:** the critical temptation $T \approx 1.8$ where complex spatial cooperation-defection dynamics emerge is recovered by C without knowledge of the payoff matrix. This constitutes independent cross-domain confirmation.
- **Langton's λ [11]:** $r(\lambda, C) = -0.001$, confirming C is not rediscovering λ's structural property. λ locates the critical neighbourhood (necessary condition); C discriminates within it (sufficient condition).

9.5. Limitations and Open Questions

1. **Substrate-specific calibration.** Several metric peaks remain empirically located on 1D. An independent theoretical derivation of why complex systems occupy these specific regimes would substantially strengthen the universality claim.
2. **P6 partial operationalisation.** The spatial PD validates P6 in a discrete spatial game substrate. Operationalising P6 in continuous substrates (N-body, EEG) remains open; the current evidence suggests this gap may explain the N-body ranking anomaly (Section 9).
3. **PD generalisation.** The current PD experiment uses the Nowak-May deterministic update rule on a square grid. Other spatial topologies, stochastic update rules, or larger neighbourhoods may shift the critical T value. Testing robustness across these variations would strengthen the claim.
4. **EEG ordering.** The top-three brain state ordering disagrees with Tononi's Φ ordering. Applying both frameworks to the same real EEG data would clarify whether this is a genuine theoretical disagreement or an artefact of synthetic signal construction.
5. **N-body cosmological extension.** The toy Lennard-Jones force law cannot reproduce full fine-tuning constraints; results should be treated as a methodological demonstration only.
6. **The candidate properties themselves.** The eight properties were derived through philosophical reasoning. Experts may identify properties that are missing, redundant, or imprecisely stated.

10. Conclusion

We have presented eight candidate properties of complexity, five measurable metrics derived from those properties, and empirical results showing that the composite correctly identifies computationally universal behaviour across five distinct substrates.

The central theoretical advance is the recognition that P1 (opaque hierarchical layering) manifests along two orthogonal axes — across spatial scale and across time — and that both should be measured. The temporal opacity metric implements this without empirical

fitting, deriving its boundaries from information theory alone. The additive combination of spatial and temporal opacity reflects the correct logical claim: satisfying one direction of P1 is sufficient evidence of hierarchical layering.

The fractal dimension metric implements P8 (scale invariance) in a way that activates naturally with substrate dimensionality. In 1D it contributes near-zero weight; in 2D it provides strong discrimination; in 3D substrates it would contribute more strongly. This is the dimensional scaling behaviour originally hypothesised.

The spatial Prisoner's Dilemma experiment adds two new contributions. First, it constitutes a direct experimental validation of P6 (co-evolutionary dynamics): C peaks precisely when and because mutual evolutionary pressure between co-adapting populations is active, and three control games confirm this specificity. Second, the framework independently recovers the Nowak-May critical temptation $T \approx 1.8$ from dynamical measurements alone, providing cross-domain convergence with evolutionary game theory. This mirrors the N-body result's convergence with physical fine-tuning arguments, and strengthens the central claim: the composite detects a substrate-agnostic property of dynamical systems that independent scientific traditions have each been measuring from the outside.

The most significant open question is whether the remaining empirically-located parameters have theoretical derivations. If so, the framework becomes predictive. If not, it remains a useful and reproducible measurement procedure with demonstrated cross-substrate consistency.

We invite experts in complexity science, dynamical systems, information theory, and computational physics to evaluate the methodology, identify errors, and propose improvements. All code and data are available at https://github.com/camerontbelt/complexity-framework.

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A. Candidate Properties — Full Statement

P1 — Opaque Hierarchical Layering: $H(S_K \mid S_{K+1}) > \varepsilon$ and $H(S_{K+1} \mid S_K) > \varepsilon$ for all adjacent layer pairs. Complexity of degree N requires N such opaque boundaries. Measurable in space (two directions) and time.

P2 — Modular Interconnection: Modules are internally coherent and externally coupled. Optimal topology is scale-free small-world. Complexity scales multiplicatively with modular richness at each layer.

P3 — Self-Assembly Without Top-Down Specification: Complexity cannot be engineered; it must self-assemble. The moment an interface is fully specified it ceases to be genuinely opaque.

P4 — Recursive Self-Modification: Maximum complexity is bounded by capacity for self-modification. Fixed-rule systems explore but cannot expand their state space.

P5 — Selection Pressure: Asymmetric environmental constraints are required for directed complexity rather than drift.

P6 — Co-evolutionary Dynamics: Maximum complexity is determined by richness of external relationships. Complexity is fundamentally ecological rather than individual.

P7 — Thermodynamic Drive: Complexity is an expression of entropy maximisation, not opposed to it. Dissipative structures are thermodynamically favoured.

P8 — Scale Invariance (meta-property): The properties apply at every scale. The universe is an instance of the phenomenon these properties describe.

B. Metric Parameters

Table 8: All metric parameters used in experiments. tcomp attractor peaks are dynamically confirmed values, not arbitrary fitting points.

Parameter	Value	Status	Substrate
Entropy tanh k	50	Parameter-free	All
Entropy var. bonus μ_σ	0.012	Fitted: C4 cluster, 1D	All
Entropy var. bonus σ_σ	0.008	Fitted: C4 cluster, 1D	All
Spatial opacity↑ peak	0.14	Fitted: C4 cluster, 1D	1D CA
Spatial opacity↑ σ	0.10	Fitted: C4 cluster, 1D	1D CA
Spatial opacity↓ peak	0.97	Fitted: C4 cluster, 1D	1D CA
Spatial opacity↓ σ	0.05	Fitted: C4 cluster, 1D	1D CA
Spatial opacity tanh k	5	Parameter-free	Non-CA
Temporal opacity tanh k	10	Parameter-free	All
tcomp peak 1 / σ	0.58 / 0.08	Confirmed: 1D C4 rand. IC	1D, 2D
tcomp peak 2 / σ	0.73 / 0.08	Confirmed: 1D C4 sc. IC	1D
tcomp peak 3 / σ	0.90 / 0.05	Confirmed: 2D C4 attractor	2D
Gzip peak / σ	0.10 / 0.05	Fitted: C4 cluster, 1D	All
Fractal dim. excess peak	0.35	Fitted: 2D C4 cluster	2D+
Fractal dim. excess σ	0.20	Fitted: 2D C4 cluster	2D+

C. Experimental Results Summary

Table 9: Summary of results across all experiments.

Experiment	n	F1	d	$\lvert r \rvert$	Notes
1D ECA (v5)	256	1.000	29.6	1.000	All 4 C4 ranks 1–4
2D Life (v5)	17	1.000	9.64	1.000	All 4 C4 ranks 1–4
2D Rule Survey	1000	—	—	—	C4 rules rank 1–3 of 1007
k=3 CA	32	0.000	—	—	Mislabelled ground truth
N-body (v4)	256	0.959	3.89	0.982	Our universe: 54th percentile
EEG synthetic	5	—	—	—	Seizure & active task correctly lowest
Spatial PD	21	—	—	—	Peak $C = 1.06$ at $T = 1.80$; matches Nowak-May
PD controls	3	—	—	—	Stag Hunt, Minority game: $C \approx 0$

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